Problem No : 1

Problem Statement :

Create a university database with tables for students, courses, and enrollments. Define primary and foreign keys to link them, insert sample data, and perform basic queries like update, delete, and select.

Source Code and Output:

File Edit View Query Database Server Tools Scripting Help

SQL File 6* SQL File 7*

Limit to 1000 rows

```sql
DROP DATABASE IF EXISTS university;
CREATE DATABASE university;
USE university;
DROP TABLE IF EXISTS students;
CREATE TABLE students(
	id INT PRIMARY KEY,
	name VARCHAR(100),
	dob DATE,
	dept VARCHAR(200)
);
INSERT INTO students
VALUES
(1,'tourist','2007-05-10','CSE'),
(2,'jiangly','2005-04-11','CSE'),
(3,'Petr','2000-04-11','CSE'),
(4,'Benq','1999-04-07','CSE'),
(5,'Errichto','2000-06-12','CSE');

SELECT * FROM students;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

id	name	dob	dept
1	tourist	2007-05-10	CSE
2	jiangly	2005-04-11	CSE
3	Petr	2000-04-11	CSE
4	Benq	1999-04-07	CSE
5	Errichto	2000-06-12	CSE

```sql
DROP TABLE IF EXISTS courses;
CREATE TABLE courses(
    course_id INT PRIMARY KEY,
    title VARCHAR(100),
    credit INT
);

INSERT INTO courses
VALUES
(101, 'DBMS', 3),
(102, 'OS', 3),
(103, 'ALGO', 3),
(104, 'DS', 3),
(105, 'OOP', 3);

SELECT *
FROM courses;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

course_id	title	credit
101	DBMS	3
102	OS	3
103	ALGO	3
104	DS	3
105	OOP	3

```sql
DROP TABLE IF EXISTS enrollment;
CREATE TABLE enrollment (
    student_id INT,
    course_id INT,
    grade CHAR(2),
    PRIMARY KEY (student_id, course_id),
    FOREIGN KEY (student_id) REFERENCES students(id),
    FOREIGN KEY (course_id) REFERENCES courses(course_id)
);

INSERT INTO enrollment
VALUES
(1,101,'A'),
(2,102,'A'),
(3,103,'A'),
(4,104,'A'),
(5,105,'A');

SELECT * FROM enrollment;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

student_id	course_id	grade
1	101	A
2	102	A
3	103	A
4	104	A
5	105	A

```sql
ALTER TABLE courses
ADD instructor VARCHAR(100);

SELECT * FROM courses;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

course_id	title	credit	instructor
101	DBMS	3	NULL
102	OS	3	NULL
103	ALGO	3	NULL
104	DS	3	NULL
105	OOP	3	NULL

```sql
UPDATE students
SET name = 'Alice Jon'
WHERE id = 1;

SELECT * FROM students;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

id	name	dob	dept
1	Alice Jon	2007-05-10	CSE
2	jiangly	2005-04-11	CSE
3	Petr	2000-04-11	CSE
4	Benq	1999-04-07	CSE
5	Errichto	2000-06-12	CSE

```sql
SELECT name, dob
FROM students
WHERE YEAR(dob) > 2000;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

name	dob
Alice Jon	2007-05-10
jiangly	2005-04-11

```sql
DELETE
FROM enrollment
WHERE course_id = 101;

DELETE
FROM courses
WHERE course_id = 101;

SELECT * FROM courses;
```

Result Grid | Filter Rows:

course_id	title	credit	instructor
102	OS	3	NULL
103	ALGO	3	NULL
104	DS	3	NULL
105	OOP	3	NULL

```sql
SELECT * FROM enrollment;
```

Result Grid | Filter Rows:

student_id	course_id	grade
2	102	A
3	103	A
4	104	A
5	105	A
NULL	NULL	NULL